

Exceptional collections for canonical stacks of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities

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Abstract

We study derived categories associated with log del Pezzo surfaces whose singularities are of type $\frac{1}{3}(1, 1)$. The Ishii–Ueda special McKay correspondence and the rationality of the minimal resolution provide the structural mechanism for full exceptional collections on the canonical stack. We make the residual-gerbe contribution explicit, obtain a length formula, and use the Corti–Heuberger cascades to give a uniform construction procedure once a full exceptional collection on the resolution of the rigid base is fixed. For the sporadic Johnson–Kollár hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$, we use the Reid–Suzuki realization as the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points to construct an explicit full exceptional collection of length 13. We compute its Euler form and further record a closed block formula in the one- $\frac{1}{n}(1, 1)$ case with blow-ups at distinct smooth points. We also obtain a semiorthogonal decomposition of the singular surface with a $K(3, 1)$ -component and ten exceptional objects via Karmazyn–Kuznetsov–Shinder descent. Finally, an appendix determines the Brauer group throughout the six Corti–Heuberger cascades: it is trivial in the cascades with $1 \leq k \leq 5$ and is isomorphic to $\mathbb{Z}/3$ in the six-point cascade.

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1 Introduction

Let X be a log del Pezzo surface over $\mathbb{C}$. Thus X is a normal projective surface with klt singularities and ample anticanonical divisor $-K_X$. In dimension two, klt singularities are precisely quotient singularities. In this paper we focus on the case where all singularities are cyclic quotient singularities of type $\frac{1}{3}(1, 1)$.

We study two related derived categories attached to X . The first is the derived category of the canonical smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$. This stack is isomorphic to X over the smooth locus and records the stabilizer groups at the quotient singularities. The second is the derived category $D^b(\text{coh } X)$ of the singular surface itself. These categories reflect different geometries: $\mathcal{X}$ is smooth as a stack, whereas X is singular as a variety.

This distinction is natural for log del Pezzo surfaces in weighted projective spaces. Johnson–Kollár classified anticanonically embedded quasismooth log del Pezzo hypersurfaces in weighted projective 3-spaces [13]. Later work on exceptional collections in this direction includes Elagin’s construction for a family of non-toric log-terminal del Pezzo hypersurfaces, treated as smooth stacks [8], and the work of Gugiatti–Rota on full exceptional collections for canonical stacks associated with the main Johnson–Kollár series [9].

The abstract existence of a full exceptional collection on $D^b(\mathrm{coh} \mathcal{X})$ follows by combining the Ishii–Ueda special McKay correspondence with the rationality of the minimal resolution; see [12, Theorem 1.4] and, for example, [9, Remark 4.8]. More precisely, Ishii–Ueda construct a fully faithful functor

$$\Phi : D^b(\mathrm{coh} \tilde{X}) \longrightarrow D^b(\mathrm{coh} \mathcal{X}),$$

and identify its semiorthogonal complement in terms of the non-special representations of the local stabilizer groups [12, Proposition 1.1 and Theorem 1.2]. For a singular point p of type $\frac{1}{3}(1, 1)$, the unique non-special representation of μ_3 is ρ_2 ; see [12, Theorem 3.2] and [9, Example 2.6]. If $\iota_p : B\mu_3 \hookrightarrow \mathcal{X}$ is the residual gerbe over p , we set

$$\mathcal{E}_p := \iota_{p*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

For singularities of type $\frac{1}{3}(1, 1)$, we make the local contribution explicit and compute the length of the resulting collection: each stacky point contributes the single exceptional object $\mathcal{E}_p$ supported on its residual gerbe. This structural construction is the starting point for the more explicit geometric and categorical results below.

Our geometric tool is the Corti–Heuberg classification by log del Pezzo cascades [5]. In this classification, the relevant surfaces are organized by sequences of blow-ups starting from $\mathbb{Q}$ -Gorenstein rigid (qG -rigid) base surfaces. Once a full exceptional collection is fixed on the minimal resolution of a rigid base, Orlov’s blow-up formula and the local Ishii–Ueda contribution give a uniform construction procedure along the corresponding cascade. Thus the cascades control both the birational geometry and the passage from the chosen base collection to the stack-theoretic collections.

Specializing this formal mechanism to $\frac{1}{3}(1, 1)$ -points gives the following explicit description and length formula.

Theorem 1.1. *Let X be a log del Pezzo surface whose singularities are $p_1, \dots, p_r$, all of type $\frac{1}{3}(1, 1)$. Let $\mathcal{X}$ be its canonical smooth Deligne–Mumford stack, and let $\tilde{X} \rightarrow X$ be the minimal resolution. The category $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection of length*

$$\ell = 12 - K_X^2 + \frac{4r}{3}.$$

More precisely, let

$$\langle G_1, \dots, G_N \rangle$$

be a full exceptional collection on $D^b(\mathrm{coh} \tilde{X})$, where

$$N = \mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3}.$$

If $\Phi : D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$ is the Ishii–Ueda fully faithful functor, the ordered collection

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(G_1), \dots, \Phi(G_N) \rangle$$

is a full exceptional collection on $D^b(\mathrm{coh} \mathcal{X})$, where

$$\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2), \quad \iota_{p_i} : B\mu_3 \hookrightarrow \mathcal{X}$$

is the inclusion of the residual gerbe over p_i , and ρ_2 is the unique non-special representation of μ_3 .

The theorem is established in Section 3 by combining the Ishii–Ueda decomposition with the geometry of the minimal resolution. Its explicit content is the identification of the local object and the length formula: the resolution contributes $12 - K_X^2 + \frac{r}{3}$ objects, and the stack contributes one object at each singular point. The broader formal consequence for quotient singularities, the corresponding formula for singularities of type $\frac{1}{k}(1, 1)$, and a closed Euler-matrix formula in the one-point case are recorded in Section 6.

We now turn to the family of degree-10 hypersurfaces

$$X_{10} = V(F_{10}) \subset \mathbb{P}(1, 2, 3, 5).$$

A general quasismooth well-formed member has a unique singular point of the type considered above and belongs to the Corti–Heuberger family $X_{1,1/3}$. By the *one-point cascade* we mean the cascade containing surfaces with one $\frac{1}{3}(1, 1)$ -point. Corti–Heuberger show that every member of $X_{1,1/3}$ admits a presentation by eight blow-ups of nonsingular points of the qG -rigid base $\mathbb{P}(1, 1, 3)$ [5, Corollary 8(1)]. The Reid–Suzuki construction realizes the general member as the degree-10 anticanonical model of such a blow-up [18, Section 1 and Table 1.1]. On minimal resolutions, this gives eight blow-ups of $\mathbb{F}_3$, all away from its negative section.

This geometry permits an explicit treatment of X_{10} : we display a full exceptional collection of length 13 on its canonical stack, compute its Euler form, and verify the compatibility and adherence conditions needed for Karmazyn–Kuznetsov–Shinder descent to the singular surface [14]. These constitute the principal explicit results of the paper.

For a blow-up $\beta_i : Y_i \rightarrow Y_{i-1}$ with exceptional curve D_i , Orlov’s blow-up formula contributes $\mathcal{O}_{D_i}(-1)$. We denote its image after pullback through the subsequent blow-ups by $G_i \in D^b(\mathrm{coh} \tilde{X}_{10})$, and retain the notation $\mathcal{E}_p$ and Φ introduced above.

Theorem 1.2. *Let $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ be a general member of the family of degree-10 log del Pezzo hypersurfaces. Its unique singularity p is of type $\frac{1}{3}(1, 1)$, and X_{10} is the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points.*

On minimal resolutions, this gives a sequence

$$Y_8 = \tilde{X}_{10} \xrightarrow{\beta_8} Y_7 \rightarrow \cdots \rightarrow Y_1 \xrightarrow{\beta_1} Y_0 = \mathbb{F}_3,$$

where each β_i is the blow-up of a point away from the negative section $C_0 \subset \mathbb{F}_3$ and its successive strict transforms. Let f be the fiber class on $\mathbb{F}_3$, and let $\sigma : \tilde{X}_{10} \rightarrow \mathbb{F}_3$ be the composition of the β_i . The category $D^b(\mathrm{coh} X_{10})$ has the following full exceptional collection of length 13:

$$\langle \mathcal{E}_p, \Phi(G_8), \dots, \Phi(G_1), \Phi\sigma^*\mathcal{O}, \Phi\sigma^*\mathcal{O}(f), \Phi\sigma^*\mathcal{O}(C_0 + 3f), \Phi\sigma^*\mathcal{O}(C_0 + 4f) \rangle.$$

Moreover, the singular surface X_{10} admits a semiorthogonal decomposition

$$D^b(\mathrm{coh} X_{10}) = \left\langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \right\rangle,$$

where $F_1, \dots, F_{10}$ are exceptional objects and

$$K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2.$$

The geometry and categorical constructions in [Theorem 1.2](#) are developed in [Section 5](#), including the exceptional collection, its Euler form, and the KKS descent to the singular surface. A second global contribution concerns the Brauer obstruction: the vanishing for X_{10} , proved in [Section 4](#), leads to the corresponding question for all six Corti–Heuberger cascades. [Appendix A](#) answers it uniformly.

Theorem 1.3 (Brauer obstruction in the Corti–Heuberger cascades). *Let X belong to a Corti–Heuberger cascade with k singular points, all of type $\frac{1}{3}(1, 1)$. Its Brauer group is*

$$\mathrm{Br}(X) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

In particular, among the six Corti–Heuberger cascades, the six-point cascade is the unique one for which the Brauer group obstructs an untwisted adherent decomposition in the sense of Karmazyn–Kuznetsov–Shinder.

The obstruction in the six-point case concerns the untwisted adherent construction only; it does not rule out unrelated semiorthogonal decompositions or the twisted adherent constructions of Karmazyn–Kuznetsov–Shinder.

The paper is organized as follows. [Section 2](#) concerns resolutions and exceptional collections, [Section 3](#) treats canonical stacks and the special McKay correspondence, and [Section 4](#) discusses the singular surface and KKS descent. [Section 5](#) develops the Corti–Heuberger cascade and the X_{10} example, including the Euler form, while [Section 6](#) gives the general $\frac{1}{k}(1, 1)$ length formula and a closed Euler-matrix formula in the one-point case. [Appendix A](#) contains the Brauer-group calculations.

2 Resolutions and exceptional collections

We recall the two standard inputs used throughout: rationality of the minimal resolution and preservation of full exceptional collections under blow-ups. Throughout this paper, all algebraic varieties are assumed to be defined over the field $\mathbb{C}$ of complex numbers.

Most of the foundational material on log del Pezzo surfaces discussed in this section can be found in [\[1\]](#). The rationality of their minimal resolutions is a standard consequence of the minimal model program (see, for instance, a recent exposition in [\[9, Remark 4.3\]](#)). Furthermore, our treatment of exceptional sequences on smooth rational surfaces closely follows the perspective of [\[10\]](#).

Definition 2.1. A log del Pezzo surface is a normal projective surface X with klt singularities such that $-K_X$ is ample.

For normal surfaces, klt singularities are precisely quotient singularities [\[15, Corollary 5.21\]](#). Moreover, klt surface singularities are rational [\[15, Theorem 5.22\]](#). Hence, if $f : \tilde{X} \rightarrow X$ is any resolution, we have $f_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^i f_*\mathcal{O}_{\tilde{X}} = 0$ for all $i > 0$. Consequently, the Leray spectral sequence gives a canonical isomorphism

$$H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) \cong H^1(X, \mathcal{O}_X).$$

Although the rationality statement is standard (see, for example, [\[1, Lemma 1.3\]](#)), we include the following short self-contained argument because it also fixes the intersection-theoretic conventions used below.

Proposition 2.2. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The surface $\tilde{X}$ is smooth and rational. In particular, X is rational.*

Proof. Since X has klt singularities and $-K_X$ is ample, Kawamata–Viehweg vanishing gives $H^1(X, \mathcal{O}_X) = 0$. Since the singularities of X are rational, the Leray isomorphism above gives $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$.

It remains to show that $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$ holds. Suppose for contradiction that there exists an effective divisor $D \sim 2K_{\tilde{X}}$. The pullback $-f^*K_X$ is nef and big, because $-K_X$ is ample. Hence $(-f^*K_X) \cdot D \geq 0$. On the other hand, if

$$K_{\tilde{X}} = f^*K_X + \sum_i a_i E_i$$

is the discrepancy formula, then we have $f^*K_X \cdot E_i = 0$ for every exceptional curve E_i . Therefore, we obtain

$$(-f^*K_X) \cdot D = (-f^*K_X) \cdot 2K_{\tilde{X}} = -2K_X^2 < 0.$$

This contradiction implies $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Thus, by Castelnuovo’s rationality criterion, $\tilde{X}$ is rational. Since X is birational to $\tilde{X}$, the surface X is rational as well. $\square$

Example 2.3. The weighted projective plane $\mathbb{P}(1, 1, 3)$ is a log del Pezzo surface with a unique singularity of type $\frac{1}{3}(1, 1)$. Its minimal resolution is the Hirzebruch surface $\mathbb{F}_3 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(3))$, and the exceptional curve is the negative section $C_0 \subset \mathbb{F}_3$ with $C_0^2 = -3$; see [14, Example 3.17], or the toric resolution procedure in [6, Chapters 10–11].

We now recall why every smooth projective rational surface admits a full exceptional collection.

Definition 2.4. Let $\mathcal{T}$ be a triangulated category. An object $E \in \mathcal{T}$ is *exceptional* if $\text{Hom}(E, E) = \mathbb{C}$ and $\text{Ext}^k(E, E) = 0$ for all $k \neq 0$. A sequence $\langle E_1, \dots, E_n \rangle$ is an *exceptional collection* if each object E_i is exceptional and $\text{Ext}^\bullet(E_j, E_i) = 0$ for all $j > i$. The collection is *full* if it generates the triangulated category.

The projective line has the standard full exceptional collection

$$D^b(\text{coh } \mathbb{P}^1) = \langle \mathcal{O}, \mathcal{O}(1) \rangle,$$

and the projective plane has Beilinson’s full exceptional collection [2]

$$D^b(\text{coh } \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

Applying Orlov’s projective bundle formula [17, Corollary 2.7] to

$$\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(n))$$

and using the collection on $\mathbb{P}^1$, we obtain a full exceptional collection on $\mathbb{F}_n$. Let C_0 denote the negative section and let f denote the fiber class. With the convention $C_0^2 = -n$, $C_0 \cdot f = 1$, and $f^2 = 0$, the relative tautological line bundle is

$$\mathcal{O}_{\mathbb{F}_n}(1) \simeq \mathcal{O}(C_0 + nf).$$

Thus one such collection is

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle.$$

Theorem 2.5 (Orlov’s blow-up formula [17, Theorem 4.3 and Corollary 4.4]). *Let S be a smooth projective surface, and let $\beta : \mathrm{Bl}_p S \rightarrow S$ be the blow-up of S at a point p , with exceptional divisor E . There is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} \mathrm{Bl}_p S) = \langle \mathcal{O}_E(-1), \beta^* D^b(\mathrm{coh} S) \rangle,$$

where $\mathcal{O}_E(-1)$ is regarded as an object of $D^b(\mathrm{coh} \mathrm{Bl}_p S)$ via the inclusion $E \hookrightarrow \mathrm{Bl}_p S$. In particular, if $D^b(\mathrm{coh} S)$ admits a full exceptional collection, the category $D^b(\mathrm{coh} \mathrm{Bl}_p S)$ also admits one.

Corollary 2.6 ([10, Theorem 5.6]). *Every smooth projective rational surface admits a full exceptional collection.*

Proof. Every smooth projective rational surface is obtained from either $\mathbb{P}^2$ or a Hirzebruch surface $\mathbb{F}_n$ by a sequence of blow-ups at points. The surfaces $\mathbb{P}^2$ and $\mathbb{F}_n$ have full exceptional collections. By Theorem 2.5, each point blow-up preserves the existence of a full exceptional collection. Hence every smooth projective rational surface admits a full exceptional collection. $\square$

Combining Proposition 2.2 and Corollary 2.6, we obtain the following consequence.

Corollary 2.7. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The category $D^b(\mathrm{coh} \tilde{X})$ admits a full exceptional collection.*

3 Canonical stacks and exceptional collections

We now introduce the stack-theoretic replacement of a singular surface and recall the local representation-theoretic input needed for the semiorthogonal decomposition. We then prove the full exceptional collection theorem for canonical stacks of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities.

Definition 3.1. Let X be a normal surface with quotient singularities. The *canonical stack* of X is the smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$ whose coarse moduli space is X , whose local models at quotient singularities are the corresponding quotient stacks, and whose structure morphism is an isomorphism over the smooth locus of X .

At a singular point of type $\frac{1}{3}(1, 1)$, the local analytic model is $\mathbb{A}^2/\mu_3$, where μ_3 acts by $\zeta \cdot (x, y) = (\zeta x, \zeta y)$. Hence the canonical stack is locally $[\mathbb{A}^2/\mu_3]$. Moreover, the minimal resolution of this singularity is determined by the Hirzebruch–Jung continued fraction $\frac{3}{1} = [3]$, so the exceptional locus consists of one smooth rational curve E with $E^2 = -3$.

We recall the local terminology used in the special McKay correspondence. Let $G \subset \mathrm{GL}_2(\mathbb{C})$ be a finite small subgroup, let $R = \mathbb{C}[x, y]$, and let

$$\tau : Y \longrightarrow \mathrm{Spec} R^G$$

be the minimal resolution of the quotient singularity. Here small means that G contains no pseudo-reflections. For an irreducible representation ρ of G , set

$$M_\rho := (R \otimes \rho^\vee)^G.$$

This is a reflexive R^G -module. The associated *full sheaf* on Y is

$$\mathcal{M}_\rho := \tau^* M_\rho / \text{torsion}.$$

Wunram's correspondence associates indecomposable full sheaves on Y with irreducible representations of G . A full sheaf $\mathcal{M}_\rho$ is called *special* if

$$H^1(Y, \mathcal{M}_\rho^\vee) = 0.$$

Equivalently, the corresponding representation ρ is called a *special representation*. A representation is called *non-special* if it is not special; see [19] and also [12, Section 2].

Lemma 3.2. *For a cyclic quotient singularity of type $\frac{1}{3}(1, 1)$, the unique non-special representation of the stabilizer group μ_3 is ρ_2 , where $\rho_i(\zeta) = \zeta^i$.*

Proof. In the cyclic case $\frac{1}{n}(1, q)$, the special representations can be read from the sequence associated with the Hirzebruch–Jung continued fraction; see [12, Theorem 3.2]. In our case, $n = 3$, $q = 1$, and $\frac{3}{1} = [3]$, so the associated sequence is

$$i_0 = 3, \quad i_1 = 1, \quad i_2 = 0.$$

Character indices are read modulo 3. Hence the special representations satisfy

$$\rho_{i_0} = \rho_3 = \rho_0, \quad \rho_{i_1} = \rho_1$$

are the special representations. Since the irreducible characters of μ_3 are exactly ρ_0, ρ_1, ρ_2 , the unique non-special representation is ρ_2 . This local computation is also noted in [9, Example 2.6]. $\square$

Definition 3.3. Let $p \in X$ be a singular point of type $\frac{1}{3}(1, 1)$, and let

$$\iota_p : B\mu_3 \hookrightarrow \mathcal{X}$$

be the residual gerbe over p . The *local exceptional object* associated to p is defined as

$$\mathcal{E}_p := \iota_{p*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

Equivalently, after identifying an étale neighborhood of p with the quotient stack $[\mathbb{A}^2/\mu_3]$, the object $\mathcal{E}_p$ is represented by the μ_3 -equivariant module

$$\mathbb{C}[x, y]/(x, y) \otimes \rho_2.$$

In the local notation of Ishii–Ueda, this corresponds to the object $\mathcal{O}_0 \otimes \rho_2$.

Theorem 3.4 (Ishii–Ueda [12, Theorem 1.4]). *Let X be a surface with at worst quotient singularities, let $\mathcal{X}$ be its canonical stack, and let $Y \rightarrow X$ be the minimal resolution. There is a fully faithful functor*

$$\Phi : D^b(\text{coh } Y) \rightarrow D^b(\text{coh } \mathcal{X})$$

and a semiorthogonal decomposition

$$D^b(\text{coh } \mathcal{X}) = \left\langle \mathcal{E}_1, \dots, \mathcal{E}_\ell, \Phi(D^b(\text{coh } Y)) \right\rangle,$$

where $\mathcal{E}_1, \dots, \mathcal{E}_\ell$ form an exceptional collection.

For the cyclic singularities $\frac{1}{n}(1, 1)$ used below, the exceptional collection in the Ishii–Ueda complement is given by the residual-gerbe simple objects corresponding to the non-special representations; see [12, Proposition 1.1 and Theorem 1.2] and [9, Example 2.6]. Hence a singularity of type $\frac{1}{3}(1, 1)$ contributes exactly one exceptional object to the complement, namely the object $\mathcal{E}_p$ defined above.

We now prove the main theorem for canonical stacks.

Theorem 3.5. *Let X be a log del Pezzo surface whose singularities are all of type $\frac{1}{3}(1, 1)$. Let $\mathcal{X}$ be its canonical stack, and let $\tilde{X} \rightarrow X$ be the minimal resolution. Suppose that the singular points of X are $p_1, \dots, p_r$. There is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(D^b(\mathrm{coh} \tilde{X})) \rangle.$$

In particular, $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection.

Proof. By Proposition 2.2, the minimal resolution $\tilde{X}$ is a smooth rational surface. Therefore $D^b(\mathrm{coh} \tilde{X})$ admits a full exceptional collection by Corollary 2.6.

By Theorem 3.4, there is a fully faithful functor

$$\Phi : D^b(\mathrm{coh} \tilde{X}) \hookrightarrow D^b(\mathrm{coh} \mathcal{X})$$

and a semiorthogonal decomposition whose complement is generated by the local exceptional objects associated with the non-special representations. Since each singular point p_i has type $\frac{1}{3}(1, 1)$, its unique non-special representation is ρ_2 by Lemma 3.2. Hence the local exceptional collection over p_i consists of the single object

$$\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

Thus the decomposition becomes

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(D^b(\mathrm{coh} \tilde{X})) \rangle.$$

The objects $\mathcal{E}_{p_i}$ have disjoint supports for distinct p_i , so they are mutually orthogonal. Since $D^b(\mathrm{coh} \tilde{X})$ has a full exceptional collection and the remaining components are generated by the exceptional objects $\mathcal{E}_{p_i}$, the category $D^b(\mathrm{coh} \mathcal{X})$ has a full exceptional collection. $\square$

Corollary 3.6. *With the notation of Theorem 3.5, the category $D^b(\mathrm{coh} \mathcal{X})$ has a full exceptional collection of length*

$$12 - K_X^2 + \frac{4r}{3}.$$

Equivalently, the minimal resolution satisfies

$$\mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3},$$

and the canonical stack adds one further exceptional object for each singular point.

Proof. Let $C_i \subset \tilde{X}$ be the exceptional curve over p_i . Since each singularity has type $\frac{1}{3}(1, 1)$, we have $C_i^2 = -3$, and the discrepancy formula is

$$K_{\tilde{X}} = f^* K_X - \frac{1}{3} \sum_{i=1}^r C_i.$$

The curves C_i are disjoint, and $f^*K_X \cdot C_i = 0$ for all i . It follows that

$$K_{\tilde{X}}^2 = K_X^2 + \frac{1}{9} \sum_{i=1}^r C_i^2 = K_X^2 - \frac{r}{3}.$$

Since $\tilde{X}$ is a smooth rational surface, we have $\rho(\tilde{X}) = 10 - K_X^2$. Therefore, the Grothendieck-group rank is

$$\mathrm{rk} K_0(\tilde{X}) = 2 + \rho(\tilde{X}) = 12 - K_X^2 = 12 - K_X^2 + \frac{r}{3}.$$

Finally, the semiorthogonal decomposition in [Theorem 3.5](#) adds one exceptional object $\mathcal{E}_{p_i}$ for each singular point p_i . Thus the full exceptional collection on $D^b(\mathrm{coh} \mathcal{X})$ has length

$$\left(12 - K_X^2 + \frac{r}{3}\right) + r = 12 - K_X^2 + \frac{4r}{3}. \quad \square$$

4 The singular surface and the KKS approach

We now turn from the canonical stack to the singular surface X . The categories $D^b(\mathrm{coh} \mathcal{X})$ and $D^b(\mathrm{coh} X)$ should not be confused: the stack $\mathcal{X}$ is smooth, whereas its coarse moduli space X is singular. For $D^b(\mathrm{coh} X)$, we use the approach of Karmazyn–Kuznetsov–Shinder to derived categories of surfaces with rational singularities.

Let $\tau : \tilde{X} \rightarrow X$ be a resolution. The method of Karmazyn–Kuznetsov–Shinder starts with a semiorthogonal decomposition of $D^b(\mathrm{coh} \tilde{X})$ and asks when it descends to a semiorthogonal decomposition of $D^b(\mathrm{coh} X)$. The relevant compatibility condition requires the objects $\mathcal{O}_E(-1)$, where E runs through the irreducible exceptional curves of τ , to belong to the appropriate components of the semiorthogonal decomposition on $\tilde{X}$; see [\[14, Definition 2.7 and Theorem 2.12\]](#). Under this condition, the pushforward functor induces a semiorthogonal decomposition of $D^b(\mathrm{coh} X)$.

The basic model for us is the contraction

$$\tau : \mathbb{F}_3 \longrightarrow \mathbb{P}(1, 1, 3),$$

which contracts the negative section $C_0 \subset \mathbb{F}_3$. We use the notation of [Section 2](#):

$$C_0^2 = -3, \quad f^2 = 0, \quad C_0 \cdot f = 1,$$

where f is the class of a fiber of the ruling $\mathbb{F}_3 \rightarrow \mathbb{P}^1$. The section disjoint from C_0 is linearly equivalent to $C_0 + 3f$.

Karmazyn–Kuznetsov–Shinder treat the contraction

$$\mathbb{F}_d \longrightarrow \mathbb{P}(1, 1, d)$$

using a full exceptional collection on $\mathbb{F}_d$. In their notation, E is the exceptional curve, H is the pullback of the point class from $\mathbb{P}^1$, and C is the section disjoint from E . The collection is

$$D^b(\mathrm{coh} \mathbb{F}_d) = \langle \mathcal{O}(-H - E), \mathcal{O}(-H), \mathcal{O}, \mathcal{O}(C) \rangle.$$

For $d = 3$, this gives the following full exceptional collection on $\mathbb{F}_3$:

$$D^b(\mathrm{coh} \mathbb{F}_3) = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + 3f) \rangle.$$

We group it as

$$\tilde{\mathcal{A}}_0 = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f) \rangle, \quad \tilde{\mathcal{A}}_1 = \langle \mathcal{O} \rangle, \quad \tilde{\mathcal{A}}_2 = \langle \mathcal{O}(C_0 + 3f) \rangle.$$

KKS show that this decomposition is compatible with τ and that its first component is untwisted adherent to C_0 in the sense of [14, Definition 3.6 and Example 3.17]. The compatibility can also be seen directly from the exact sequence

$$0 \longrightarrow \mathcal{O}(-C_0 - f) \longrightarrow \mathcal{O}(-f) \longrightarrow \mathcal{O}_{C_0}(-1) \longrightarrow 0$$

which shows that

$$\mathcal{O}_{C_0}(-1) \in \tilde{\mathcal{A}}_0.$$

Thus the compatibility condition is built into the first block of the collection.

Therefore [14, Theorem 2.12] gives

$$D^b(\text{coh } \mathbb{P}(1, 1, 3)) = \langle \mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2 \rangle,$$

where $\mathcal{A}_i$ denotes the descended component. Moreover, the components satisfy

$$\mathcal{A}_0 \simeq D^b(K(3, 1)\text{-mod}), \quad \mathcal{A}_1 \simeq D^b(\mathbb{C}), \quad \mathcal{A}_2 \simeq D^b(\mathbb{C});$$

see [14, Example 3.17]. Here $D^b(\mathbb{C})$ denotes the bounded derived category of finite-dimensional complex vector spaces. The algebra $K(3, 1)$ is the Kalck–Karmazyn algebra associated with the cyclic quotient singularity $\frac{1}{3}(1, 1)$, and in this case

$$K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2;$$

see [14, Example 3.14].

We will only need the following compatibility observation.

Lemma 4.1. *Let $\beta : S' \rightarrow S$ be the blow-up of a smooth point on a smooth surface. Let $C \subset S$ be a smooth curve, and assume that the center of β does not lie on C . If $C' \subset S'$ is the strict transform of C , there is an isomorphism in $D^b(\text{coh } S')$:*

$$\mathcal{O}_{C'}(-1) \simeq \beta^* \mathcal{O}_C(-1)$$

In particular, the inclusion $\mathcal{O}_C(-1) \in \mathcal{B} \subset D^b(\text{coh } S)$ implies that

$$\mathcal{O}_{C'}(-1) \in \beta^* \mathcal{B}.$$

Proof. Since the center of the blow-up is disjoint from C , the morphism $C' \rightarrow C$ induced by β is an isomorphism. Therefore the pullback of $\mathcal{O}_C(-1)$ is $\mathcal{O}_{C'}(-1)$, viewed as a sheaf on S' . This proves the claim. $\square$

Remark 4.2. We also record that the Brauer obstruction does not appear in the examples considered below. Here $\text{Br}(X)$ denotes the cohomological Brauer group. In the present projective setting, its torsion classes are represented by Azumaya algebras by Gabber's theorem, as used in [3, proof of Theorem 6]; thus no distinction between the two Brauer groups arises for the finite groups computed here. This is also the Brauer group entering the obstruction in [14, Corollary 4.8].

We use Bright's exact sequence for a normal surface X with rational singularities. Let $\tau : Y \rightarrow X$ be the minimal resolution, and let $\Lambda_\tau \subset \text{Pic}(Y)$ denote the subgroup generated by the exceptional curves. The intersection pairing induces a homomorphism

$$\theta : \text{Pic}(Y) \longrightarrow \Lambda_\tau^\vee, \quad D \longmapsto (E \mapsto D \cdot E),$$

and there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(Y) \xrightarrow{\theta} \Lambda_\tau^\vee \longrightarrow \mathrm{Br}(X) \longrightarrow \mathrm{Br}(Y);$$

see [3, Proposition 1]. Since the resolutions considered here are smooth rational surfaces, their Brauer groups vanish.

For $X = \mathbb{P}(1, 1, 3)$, the minimal resolution is $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$, and the exceptional curve is C_0 . The exceptional lattice is $\Lambda_\tau = \mathbb{Z}C_0$, and we identify $\Lambda_\tau^\vee$ with $\mathbb{Z}$ by evaluating on C_0 . The fiber class f satisfies

$$f \cdot C_0 = 1.$$

Therefore the map

$$\mathrm{Pic}(\mathbb{F}_3) \longrightarrow \Lambda_\tau^\vee \cong \mathbb{Z}, \quad D \longmapsto D \cdot C_0,$$

is surjective. Since $\mathrm{Br}(\mathbb{F}_3) = 0$, Bright's exact sequence gives

$$\mathrm{Br}(\mathbb{P}(1, 1, 3)) = 0.$$

The same computation applies to X_{10} after the geometric identification in Section 5. Let $\nu : Y \rightarrow X_{10}$ be the minimal resolution. We will show that Y is obtained from $\mathbb{F}_3$ by blowing up eight smooth points away from C_0 . Let

$$\sigma : Y \longrightarrow \mathbb{F}_3$$

be the resulting morphism, and let $E \subset Y$ be the strict transform of C_0 . The curve E is the unique exceptional curve of ν , and $E^2 = -3$. Hence $\Lambda_\nu = \mathbb{Z}E$. Since the blow-up centers are away from C_0 , the pullback σ^*f satisfies

$$(\sigma^*f) \cdot E = f \cdot C_0 = 1.$$

Thus the map

$$\mathrm{Pic}(Y) \longrightarrow \Lambda_\nu^\vee \cong \mathbb{Z}, \quad D \longmapsto D \cdot E,$$

is surjective. Since Y is a smooth rational surface, $\mathrm{Br}(Y) = 0$, and Bright's exact sequence gives

$$\mathrm{Br}(X_{10}) = 0.$$

The behavior of this obstruction in the remaining Corti–Heuberger cascades is analyzed in Appendix A.

5 Corti–Heuberger cascades and the surface X_{10}

In this section we describe a uniform construction along the Corti–Heuberger cascades and then specialize to the hypersurface

$$X_{10} \subset \mathbb{P}(1, 2, 3, 5).$$

The abstract existence of a full exceptional collection on the canonical stack follows from the Ishii–Ueda decomposition and the rationality of the minimal resolution. Once a full exceptional collection on the minimal resolution of a qG -rigid base is fixed, the cascades give a uniform procedure: pull the collection through the successive blow-ups by Orlov's formula, apply Φ , and place the residual-gerbe objects prescribed by the local

Ishii–Ueda complement before its image. For X_{10} , we carry this out with the objects displayed explicitly.

Corti–Heuberger classify non-smooth del Pezzo surfaces with $\frac{1}{3}(1, 1)$ points into qG -deformation families grouped into unprojection cascades. A cascade is obtained by blowing up nonsingular points, or equivalently by reversing a sequence of contractions of floating (-1) -curves; see [5, Definition 5]. Since these curves lie in the smooth locus, the cascade preserves both the number and the analytic type of the singularities.

We follow the notation of [5, Theorem 4]: k denotes the number of $\frac{1}{3}(1, 1)$ points, and $d = K_X^2$ denotes the anticanonical degree of a surface X . In the Fano index one case, $X_{k,d}$ denotes the qG -deformation family with k singular points and degree d . The rigid bases are identified in [5, Theorem 6 and Remark 7], and [5, Corollary 8] organizes the remaining families into cascades of blow-ups from them. We choose a representative of each base family and denote it by Z_k :

k	Z_k	$K_{Z_k}^2$
1	$S_{1,25/3} = \mathbb{P}(1, 1, 3)$	$25/3$
2	$X_{2,17/3}$	$17/3$
3	$X_{3,5}$	5
4	$X_{4,7/3}$	$7/3$
5	$X_{5,5/3}$	$5/3$
6	$X_{6,2}$	2.

Every surface in an index-one family $X_{k,d}$ of the corresponding cascade is obtained from its listed base by blowing up nonsingular points. The base step is therefore to choose a full exceptional collection on each minimal resolution $\tilde{Z}_k$; the remaining pieces are the Orlov exceptional objects from the smooth blow-ups and the Ishii–Ueda residual-gerbe objects over the stacky points.

The auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ may appear as outputs of a directed MMP; see [5, Theorem 6 and Remark 7]. The alternative contraction routes relating smooth-point blow-ups of these families to the chosen bases are described in [5, Remark 9]; the one-point case relevant to X_{10} is made explicit below.

We now describe the inductive step. Let X be a surface in the family $X_{k,d}$. The surface X is obtained from Z_k by

$$K_{Z_k}^2 - d$$

blow-ups at nonsingular points. Passing to minimal resolutions gives a sequence of ordinary smooth blow-ups starting from $\tilde{Z}_k$. At each step, Orlov’s blow-up formula adds the exceptional object associated with the new exceptional curve. If $\langle G_1, \dots, G_N \rangle$ is the resulting collection on the resolution, the Ishii–Ueda order is

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(G_1), \dots, \Phi(G_N) \rangle,$$

where $\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2)$. This construction gives a full exceptional collection on the canonical stack of X , with length

$$\ell = 12 - K_X^2 + \frac{4k}{3}.$$

We now specialize to X_{10} . Let x_0, x_1, x_2, x_3 be the weighted coordinates on $\mathbb{P}(1, 2, 3, 5)$, with weights 1, 2, 3, 5.

Proposition 5.1. *A general hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ of weighted degree 10 is quasismooth. Moreover, it has a unique singular point,*

$$p = [0 : 0 : 1 : 0],$$

and this singularity is of type $\frac{1}{3}(1, 1)$.

Proof. We use the standard quasismoothness criterion for weighted hypersurfaces as in [11]. The base locus of $|\mathcal{O}_{\mathbb{P}(1,2,3,5)}(10)|$ is the single point $p = [0 : 0 : 1 : 0]$. On the smooth locus of the ambient weighted projective space away from p , Bertini's theorem therefore gives smoothness of a general member.

Since $\mathbb{P}(1, 2, 3, 5)$ is well formed and its weights are pairwise coprime, its singular locus consists of the three coordinate points of weights 2, 3, 5. The monomials x_1^5 and x_3^2 show that a general hypersurface does not pass through $[0 : 1 : 0 : 0]$ or $[0 : 0 : 0 : 1]$. It remains to analyze the point p . On the chart $x_2 \neq 0$, the local quotient cover is $\mathbb{A}^3$ with coordinates corresponding to x_0, x_1, x_3 , and the stabilizer μ_3 acts with weights

$$(1, 2, 5) \equiv (1, 2, 2) \pmod{3}.$$

Since $x_0 x_2^3$ has weighted degree 10, its coefficient is nonzero in a general defining polynomial. It gives a nonzero linear term in the coordinate corresponding to x_0 , so the hypersurface is smooth on the local quotient cover and we may eliminate this coordinate. The induced μ_3 -action on the tangent coordinates has weights $(2, 2)$, which is equivalent to $(1, 1)$. Thus X_{10} is quasismooth and its unique singularity is of type $\frac{1}{3}(1, 1)$. $\square$

By weighted adjunction and the standard intersection formula for weighted projective spaces [7], we have

$$\omega_{X_{10}}^\vee = \mathcal{O}_{X_{10}}((1 + 2 + 3 + 5) - 10) = \mathcal{O}_{X_{10}}(1), \quad K_{X_{10}}^2 = \frac{10}{1 \cdot 2 \cdot 3 \cdot 5} = \frac{1}{3}.$$

Thus a general $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ is a log del Pezzo surface in the Corti–Heuberger family $X_{1,1/3}$; see [5, Theorem 4 and Table 2]. By [5, Corollary 8(1)], every surface in $X_{1,d}$ is obtained by blowing up $25/3 - d$ nonsingular points of $\mathbb{P}(1, 1, 3)$; for $d = 1/3$, this gives eight blow-ups.

A chosen directed MMP in the one-point case may instead terminate at the auxiliary family $B_{1,16/3}$. This does not conflict with the presentation above: [5, Remark 9(1)] states that a blow-up $X \rightarrow B_{1,16/3}$ of a nonsingular point admits an alternative sequence of four contractions of floating (-1) -curves from X to $\mathbb{P}(1, 1, 3)$. The available (-1) -curves therefore permit different contraction sequences, and the endpoint of one directed MMP does not obstruct the eight-blow-up presentation. For the general member, the Reid–Suzuki construction realizes this presentation with eight general smooth centers and identifies its anticanonical model.

Theorem 5.2 (Reid–Suzuki [18, Section 1 and Table 1.1]). *Let $P_1, \dots, P_8 \in \mathbb{P}(1, 1, 3)$ be eight general smooth points, and let*

$$\eta : S \longrightarrow \mathbb{P}(1, 1, 3)$$

be their blow-up. The anticanonical divisor of S is

$$-K_S = \eta^*(-K_{\mathbb{P}(1,1,3)}) - \sum_{i=1}^8 E_i,$$

where E_i is the exceptional curve over P_i . Here $-K_{\mathbb{P}(1,1,3)}$ is the anticanonical $\mathbb{Q}$ -Cartier divisor, whose associated rank-one reflexive sheaf is $\mathcal{O}_{\mathbb{P}(1,1,3)}(5)$. Moreover, the anticanonical ring has the form

$$R(S, -K_S) \cong \frac{\mathbb{C}[x, y, z, w]}{(F_{10})}, \quad \deg(x, y, z, w) = (1, 2, 3, 5),$$

where F_{10} is weighted homogeneous of degree 10. Consequently, the anticanonical model of S is a degree-10 hypersurface

$$X_{10} \subset \mathbb{P}(1, 2, 3, 5).$$

The minimal resolution of X_{10} is obtained from $\mathbb{F}_3$ by blowing up the eight corresponding points away from the negative section.

Proof. By the standard formula for weighted projective spaces, the anticanonical $\mathbb{Q}$ -Cartier divisor corresponds to the rank-one reflexive sheaf $\mathcal{O}_{\mathbb{P}(1,1,3)}(5)$, and

$$K_{\mathbb{P}(1,1,3)}^2 = \frac{25}{3}.$$

Since the points P_i are smooth, the blow-up formula gives the asserted expression for $-K_S$, and

$$K_S^2 = \frac{25}{3} - 8 = \frac{1}{3}.$$

In the cited construction, Reid–Suzuki describe the cascade obtained by blowing up general smooth points of $\mathbb{P}(1, 1, 3)$; for eight blow-ups, their Table 1.1 records the anticanonical model as

$$S_{10} \subset \mathbb{P}(1, 2, 3, 5);$$

its homogeneous coordinate ring, with the anticanonical grading, is precisely the graded ring displayed in the statement.

Finally, the minimal resolution $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ is an isomorphism over the smooth locus. Since all eight centers lie in that locus, they lift uniquely to points $q_1, \dots, q_8 \in \mathbb{F}_3$ away from the negative section C_0 . Hence the minimal resolution is the blow-up of $\mathbb{F}_3$ at these eight points. $\square$

Let

$$\sigma : \tilde{X}_{10} \longrightarrow \mathbb{F}_3$$

be the blow-down morphism of [Theorem 5.2](#), let $D_1, \dots, D_8$ be its exceptional curves, and let f be the fiber class on $\mathbb{F}_3$. Starting from the standard full exceptional collection

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + 3f), \mathcal{O}(C_0 + 4f) \rangle$$

and applying Orlov’s blow-up formula gives

$$\langle \mathcal{O}_{D_1}(-1), \dots, \mathcal{O}_{D_8}(-1), \sigma^* \mathcal{O}, \sigma^* \mathcal{O}(f), \sigma^* \mathcal{O}(C_0 + 3f), \sigma^* \mathcal{O}(C_0 + 4f) \rangle$$

on $\tilde{X}_{10}$. The curves D_i are pairwise disjoint, so their exceptional objects may be listed in any order. Hence [Theorem 1.1](#) places the local object $\mathcal{E}_p$ before the image of this collection and gives the following consequence.

Corollary 5.3. *Let $\pi : \mathcal{X}_{10} \rightarrow X_{10}$ be the canonical stack. The category $D^b(\mathrm{coh} \mathcal{X}_{10})$ has a full exceptional collection of length 13.*

More generally, the Hille–Perling toric systems on $\mathbb{F}_3$ induce the following family of full exceptional collections on $\tilde{X}_{10}$ [10, Proposition 5.2]:

$$\langle \mathcal{O}_{D_1}(-1), \dots, \mathcal{O}_{D_8}(-1), \sigma^* \mathcal{O}, \sigma^* \mathcal{O}(f), \\ \sigma^* \mathcal{O}(C_0 + (s+4)f), \sigma^* \mathcal{O}(C_0 + (s+5)f) \rangle,$$

for all $s \in \mathbb{Z}$. The underlying four-term Hille–Perling collection on $\mathbb{F}_3$ is strongly exceptional for $s \geq -1$.

Proposition 5.4 (The Euler form for $\mathcal{X}_{10}$). *Let $s \in \mathbb{Z}$ and let*

$$\mathcal{C}(s) = \langle \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^* \mathcal{O}), \Phi(\sigma^* \mathcal{O}(f)), \\ \Phi(\sigma^* \mathcal{O}(C_0 + (s+4)f)), \Phi(\sigma^* \mathcal{O}(C_0 + (s+5)f)) \rangle$$

be an exceptional collection of $D^b(\text{coh } \mathcal{X}_{10})$ associated with the Hille–Perling toric system for $\mathbb{F}_3$. The matrix of the Euler form $S_{ij}(s) = \chi(\mathcal{C}_i(s), \mathcal{C}_j(s))$ is given by

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|c|cccc} 1 & 0_{1 \times 8} & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0_{8 \times 1} & I_8 & & & & -\mathbf{1}_{8 \times 4} \\ \hline & & 1 & 2 & 2s+7 & 2s+9 \\ & & 0 & 1 & 2s+5 & 2s+7 \\ & 0_{4 \times 1} & 0 & 0 & 1 & 2 \\ & & 0 & 0 & 0 & 1 \end{array} \right).$$

Equivalently, with respect to the ordered collection $\mathcal{C}(s)$, the fully expanded matrix is

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|cccccccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & -1 & -1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 2s+7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2s+5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right).$$

Proof. Let

$$\Psi : D^b(\text{coh } \mathcal{X}_{10}) \longrightarrow D^b(\text{coh } \tilde{X}_{10})$$

be the left adjoint of the Ishii–Ueda functor

$$\Phi : D^b(\text{coh } \tilde{X}_{10}) \longrightarrow D^b(\text{coh } \mathcal{X}_{10}).$$

We first compute the entries involving the residual-gerbe object $\mathcal{E}_p$. Locally at the singular point p , the stack is $[\mathbb{A}^2/\mu_3]$, and $\mathcal{E}_p$ is represented by $\mathcal{O}_0 \otimes \rho_2$, where ρ_2 is

the unique non-special representation. We use the left-adjoint computation of Gugiatti–Rota [9, Theorem 3.1 and Example 3.5]. In the case $\frac{1}{3}(1, 1)$, the natural representation is

$$\rho_{\text{nat}} = \rho_1 \oplus \rho_1, \quad \rho_{\text{det}} = \bigwedge^2 \rho_{\text{nat}} = \rho_2.$$

The minimal resolution has a single exceptional curve E , with $E^2 = -3$, and the special representations are ρ_0 and ρ_1 . Since

$$\rho_2 \otimes \rho_{\text{det}} = \rho_2 \otimes \rho_2 = \rho_1$$

holds, the cited formula gives

$$\Psi(\mathcal{E}_p) \simeq \mathcal{O}_E(-2).$$

Equivalently, [9, Example 3.5] states directly, in its local notation, that for a cyclic quotient singularity of type $\frac{1}{n}(1, 1)$,

$$\psi(\mathcal{O}_0 \otimes \rho_{n-1}) \simeq \mathcal{O}_E(-n+1).$$

Taking $n = 3$ gives the displayed identification $\Psi(\mathcal{E}_p) \simeq \mathcal{O}_E(-2)$, with no cohomological shift. Therefore, for every $A \in D^b(\text{coh } \tilde{X}_{10})$, adjunction gives

$$\chi_{\mathcal{X}_{10}}(\mathcal{E}_p, \Phi A) = \chi_{\tilde{X}_{10}}(\mathcal{O}_E(-2), A).$$

On the other hand, the Ishii–Ueda semiorthogonal decomposition has the form

$$D^b(\text{coh } \mathcal{X}_{10}) = \langle \mathcal{E}_p, \Phi(D^b(\text{coh } \tilde{X}_{10})) \rangle,$$

and we use here the ordering of [12, Theorem 1.4], in which the exceptional complement lies to the left of $\Phi(D^b(\text{coh } \tilde{X}_{10}))$. Consequently, semiorthogonality gives

$$\chi_{\mathcal{X}_{10}}(\Phi A, \mathcal{E}_p) = 0.$$

Only the first row has to be computed.

The curve E is the strict transform of the negative section $C_0 \subset \mathbb{F}_3$, while the curves $D_1, \dots, D_8$ are disjoint from E . Hence the corresponding Euler characteristics vanish:

$$\chi_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \mathcal{O}_{D_i}(-1)) = 0$$

for all i . Now let $L = \mathcal{O}_{\mathbb{F}_3}(D)$ be a line bundle on $\mathbb{F}_3$. Since E is identified with C_0 under σ , we have

$$\deg((\sigma^* L)|_E) = D \cdot C_0.$$

For the inclusion $i : E \hookrightarrow \tilde{X}_{10}$, Grothendieck duality for an effective Cartier divisor gives

$$i^!(\sigma^* L) \simeq (\sigma^* L)|_E \otimes \mathcal{O}_E(E)[-1].$$

Since $\mathcal{O}_E(E) \simeq \mathcal{O}_E(-3)$, we obtain

$$\text{RHom}_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \sigma^* L) \simeq \text{R}\Gamma(E, \mathcal{O}_E(D \cdot C_0 - 1))[-1].$$

Therefore, we have

$$\chi_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \sigma^* L) = -\chi(\mathcal{O}_{\mathbb{P}^1}(D \cdot C_0 - 1)) = -D \cdot C_0.$$

Consider the following four line bundles:

$$\mathcal{O}, \quad \mathcal{O}(f), \quad \mathcal{O}(C_0 + (s+4)f), \quad \mathcal{O}(C_0 + (s+5)f).$$

Their intersection numbers with C_0 are

$$0, \quad 1, \quad s+1, \quad s+2.$$

This gives the first row of the matrix.

We now compute the block coming from the resolution. Since Φ is fully faithful, we have

$$\chi_{\mathcal{X}_{10}}(\Phi A, \Phi B) = \chi_{\tilde{X}_{10}}(A, B)$$

for all $A, B \in D^b(\text{coh } \tilde{X}_{10})$. The curves D_i are pairwise disjoint exceptional curves of point blow-ups, so we obtain

$$\chi(\mathcal{O}_{D_i}(-1), \mathcal{O}_{D_j}(-1)) = \delta_{ij}.$$

For a line bundle L pulled back from $\mathbb{F}_3$, its restriction satisfies $(\sigma^*L)|_{D_i} \simeq \mathcal{O}_{D_i}$. Using $D_i^2 = -1$ and the same divisor-duality computation, we get

$$\text{RHom}_{\tilde{X}_{10}}(\mathcal{O}_{D_i}(-1), \sigma^*L) \simeq \text{R}\Gamma(D_i, \mathcal{O}_{D_i})[-1],$$

which implies

$$\chi(\mathcal{O}_{D_i}(-1), \sigma^*L) = -1.$$

In the opposite direction, we have

$$\text{RHom}_{\tilde{X}_{10}}(\sigma^*L, \mathcal{O}_{D_i}(-1)) \simeq \text{R}\Gamma(D_i, \mathcal{O}_{D_i}(-1)),$$

so we get

$$\chi(\sigma^*L, \mathcal{O}_{D_i}(-1)) = 0.$$

This gives the blocks I_8 , $-\mathbf{1}_{8 \times 4}$, and $\mathbf{0}_{4 \times 8}$.

It remains to compute the 4×4 block coming from $\mathbb{F}_3$. We use

$$C_0^2 = -3, \quad C_0 \cdot f = 1, \quad f^2 = 0, \quad K_{\mathbb{F}_3} = -2C_0 - 5f.$$

For line bundles $L_1 = \mathcal{O}(A_1)$ and $L_2 = \mathcal{O}(A_2)$ on $\mathbb{F}_3$, Riemann–Roch gives

$$\chi(L_1, L_2) = \chi(\mathcal{O}(A_2 - A_1)) = 1 + \frac{1}{2}(A_2 - A_1) \cdot (A_2 - A_1 - K_{\mathbb{F}_3}).$$

Applying this to

$$\mathcal{O}, \quad \mathcal{O}(f), \quad \mathcal{O}(C_0 + (s+4)f), \quad \mathcal{O}(C_0 + (s+5)f),$$

we obtain

$$\begin{pmatrix} 1 & 2 & 2s+7 & 2s+9 \\ 0 & 1 & 2s+5 & 2s+7 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Putting the four blocks together gives the stated matrix. $\square$

We finally apply the KKS descent construction of [Section 4](#) to the singular surface X_{10} ; the required compatibility after the eight blow-ups is verified in the proof below.

Theorem 5.5. *The singular surface X_{10} admits a semiorthogonal decomposition*

$$D^b(\mathrm{coh} X_{10}) = \left\langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \right\rangle,$$

where $F_1, \dots, F_{10}$ are exceptional objects and

$$K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2.$$

Proof. We start with the decomposition on $\mathbb{F}_3$ from [\[14, Example 3.17\]](#):

$$D^b(\mathrm{coh} \mathbb{F}_3) = \left\langle \tilde{\mathcal{A}}_0, \tilde{\mathcal{A}}_1, \tilde{\mathcal{A}}_2 \right\rangle,$$

whose three components are

$$\tilde{\mathcal{A}}_0 = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f) \rangle, \quad \tilde{\mathcal{A}}_1 = \langle \mathcal{O} \rangle, \quad \tilde{\mathcal{A}}_2 = \langle \mathcal{O}(C_0 + 3f) \rangle.$$

As explained in [Section 4](#), the exact sequence

$$0 \longrightarrow \mathcal{O}(-C_0 - f) \longrightarrow \mathcal{O}(-f) \longrightarrow \mathcal{O}_{C_0}(-1) \longrightarrow 0$$

shows that $\mathcal{O}_{C_0}(-1) \in \tilde{\mathcal{A}}_0$.

By [Theorem 5.2](#), the minimal resolution $\tilde{X}_{10}$ is obtained from $\mathbb{F}_3$ by eight blow-ups whose centers are disjoint from C_0 and its successive strict transforms. Let

$$\sigma : \tilde{X}_{10} \longrightarrow \mathbb{F}_3$$

be the composition of these blow-ups, and let $E \subset \tilde{X}_{10}$ be the strict transform of C_0 . For a blow-up $\beta : S' \rightarrow S$ with exceptional curve D , the right-handed form of [Theorem 2.5](#), obtained by the standard mutation, is

$$D^b(\mathrm{coh} S') = \left\langle \beta^* D^b(\mathrm{coh} S), \mathcal{O}_D \right\rangle.$$

Iterating this form gives a semiorthogonal decomposition of $D^b(\mathrm{coh} \tilde{X}_{10})$ in which the eight Orlov exceptional components follow the three pulled-back blocks. Define the two line bundles on $\mathbb{F}_3$ by

$$L_0 = \mathcal{O}(-C_0 - f), \quad L_1 = \mathcal{O}(-f).$$

The morphism σ is an isomorphism in a neighborhood of E . Since the blow-up centers are disjoint from C_0 , the relation $L_1 = L_0(C_0)$ pulls back to

$$\sigma^* L_1 = \sigma^* L_0(E).$$

The corresponding intersection number is

$$(\sigma^* L_0) \cdot E = L_0 \cdot C_0 = (-C_0 - f) \cdot C_0 = 3 - 1 = 2.$$

Since $\tilde{X}_{10}$ is rational, its structure sheaf satisfies

$$H^\bullet(\tilde{X}_{10}, \mathcal{O}_{\tilde{X}_{10}}) \cong \mathbb{C}.$$

Together with the displayed line-bundle relations, this verifies the hypotheses and equivalent conditions of [14, Lemma 3.5 and Definition 3.6] for untwisted adherence to the single (-3) -curve E . Moreover, Lemma 4.1 gives

$$\mathcal{O}_E(-1) \in \sigma^* \tilde{\mathcal{A}}_0.$$

Since E is the only exceptional curve of the contraction

$$\nu : \tilde{X}_{10} \longrightarrow X_{10},$$

this is exactly the compatibility condition of [14, Definition 2.7].

By [14, Theorem 2.12], the decomposition descends to a semiorthogonal decomposition of $D^b(\text{coh } X_{10})$. The untwisted adherent component descending from $\sigma^* \tilde{\mathcal{A}}_0$ is equivalent to $D^b(K(3, 1)\text{-mod})$ by [14, Theorem 3.16].

The remaining components consist, in this order, of the two exceptional components descending from $\tilde{\mathcal{A}}_1$ and $\tilde{\mathcal{A}}_2$, followed by the eight Orlov components from the blow-ups. Relabeling these ten exceptional components as $F_1, \dots, F_{10}$ gives the stated semiorthogonal decomposition. $\square$

6 Further consequences

The Ishii–Ueda decomposition and rationality of the minimal resolution have the following general consequence. For singularities of type $\frac{1}{k_i}(1, 1)$, both the local contribution and the length are explicit.

Proposition 6.1. *Let X be a log del Pezzo surface, and let $\mathcal{X}$ be its canonical stack. The category $D^b(\text{coh } \mathcal{X})$ admits a full exceptional collection.*

Assume moreover that the singularities of X are $p_1, \dots, p_r$, where p_i has type $\frac{1}{k_i}(1, 1)$, with $k_i \geq 2$. The length of this full exceptional collection is

$$12 - K_X^2 + \sum_{i=1}^r \frac{2(k_i - 1)(k_i - 2)}{k_i}.$$

Equivalently, for the minimal resolution $f : \tilde{X} \rightarrow X$, we have

$$\text{rk } K_0(\tilde{X}) = 12 - K_X^2 + \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i},$$

and the canonical stack contributes $k_i - 2$ further exceptional objects over the point p_i .

Proof. By Proposition 2.2 and Corollary 2.6, the minimal resolution is a smooth rational surface with a full exceptional collection. The first assertion therefore follows from Theorem 3.4.

Under the additional hypothesis, let $C_i \subset \tilde{X}$ be the exceptional curve over p_i . Thus $C_i^2 = -k_i$, and the discrepancy formula is

$$K_{\tilde{X}} = f^* K_X - \sum_{i=1}^r \frac{k_i - 2}{k_i} C_i.$$

The curves C_i are disjoint and satisfy $f^* K_X \cdot C_i = 0$. It follows that

$$K_{\tilde{X}}^2 = K_X^2 - \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i}.$$

Consequently, the rank is

$$\mathrm{rk} K_0(\tilde{X}) = 12 - K_{\tilde{X}}^2 = 12 - K_X^2 + \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i}.$$

For a singularity of type $\frac{1}{k_i}(1, 1)$, Gugiatti–Rota identify ρ_0 and ρ_1 as the special representations and order the local exceptional collection as

$$\langle \mathcal{O}_0 \otimes \rho_2, \dots, \mathcal{O}_0 \otimes \rho_{k_i-1} \rangle.$$

See [9, Example 2.6]. In the Ishii–Ueda decomposition, this local block precedes $\Phi(D^b(\mathrm{coh} \tilde{X}))$, and blocks at distinct singular points are mutually orthogonal. Hence p_i contributes $k_i - 2$ objects. Adding these local contributions to the rank above gives

$$12 - K_X^2 + \sum_{i=1}^r \left(\frac{(k_i - 2)^2}{k_i} + k_i - 2 \right) = 12 - K_X^2 + \sum_{i=1}^r \frac{2(k_i - 1)(k_i - 2)}{k_i}. \quad \square$$

The preceding proposition determines the length of the collection. In the one-point case, when the minimal resolution is obtained from $\mathbb{F}_n$ by blowing up pairwise distinct points away from the negative section, the same construction also gives a closed formula for its Euler matrix.

Proposition 6.2 (An Euler matrix for one cyclic point). *Let $n \geq 3$ and $m \geq 0$. Suppose that the minimal resolution of a log del Pezzo surface X fits into a diagram*

$$\tilde{X} \xrightarrow{\sigma} \mathbb{F}_n, \quad \tilde{X} \xrightarrow{\nu} X,$$

where σ is the blow-up of m distinct points of $\mathbb{F}_n$, all disjoint from the negative section $C_0 \subset \mathbb{F}_n$, and ν contracts the strict transform E of C_0 to the unique singular point of X , of type $\frac{1}{n}(1, 1)$. Write $D_1, \dots, D_m$ for the exceptional curves of σ , and use the intersection conventions

$$C_0^2 = -n, \quad C_0 \cdot f = 1, \quad f^2 = 0.$$

Let $\mathcal{X}$ be the canonical stack of X , let $\Phi : D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$ be the Ishii–Ueda functor, and let $\iota : B\mu_n \hookrightarrow \mathcal{X}$ be the residual gerbe. For $2 \leq i \leq n - 1$, define

$$\mathcal{E}_i := \iota_*(\mathcal{O}_{B\mu_n} \otimes \rho_i), \quad \rho_i(\zeta) = \zeta^i.$$

For $s \in \mathbb{Z}$, the Hille–Perling collection [10, Proposition 5.2], Orlov’s blow-up formula, and the Ishii–Ueda decomposition give the ordered full exceptional collection

$$\begin{aligned} \mathcal{C}_{n,m}(s) = & \langle \mathcal{E}_2, \dots, \mathcal{E}_{n-1}, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_m}(-1)), \\ & \Phi(\sigma^* \mathcal{O}), \Phi(\sigma^* \mathcal{O}(f)), \Phi(\sigma^* \mathcal{O}(C_0 + (s + n + 1)f)), \\ & \Phi(\sigma^* \mathcal{O}(C_0 + (s + n + 2)f)) \rangle. \end{aligned}$$

The local block $T_n \in M_{n-2}(\mathbb{Z})$ is defined by

$$T_n := (\chi(\mathcal{E}_i, \mathcal{E}_j))_{2 \leq i, j \leq n-1}$$

and has entries

$$(T_n)_{ij} = \delta_{ij} - 2\delta_{j,i+1} + \delta_{j,i+2}.$$

Here the Kronecker symbols refer to the displayed integer range rather than to indices modulo n . Equivalently, with the evident truncation in the small dimensions,

$$T_n = \begin{pmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \ddots & \vdots \\ 0 & 0 & 1 & -2 & \ddots & 0 \\ \vdots & & & \ddots & \ddots & \ddots \\ 0 & \cdots & 0 & 1 & -2 & \\ 0 & \cdots & \cdots & 0 & 1 & \end{pmatrix}.$$

For $n \geq 4$, the local-to-Hirzebruch block $B_n(s) \in \text{Mat}_{(n-2) \times 4}(\mathbb{Z})$ is

$$B_n(s) = \begin{pmatrix} & 0_{(n-4) \times 4} & \\ 1 & 2 & s+2 & s+3 \\ 0 & -1 & -(s+1) & -(s+2) \end{pmatrix},$$

where the zero block is absent for $n = 4$. The penultimate and final rows are indexed by $\mathcal{E}_{n-2}$ and $\mathcal{E}_{n-1}$, respectively. For $n = 3$, the local block consists only of $\mathcal{E}_2$, so $T_3 = (1)$. The representation $\rho_{n-2} = \rho_1$ is special and does not belong to the Ishii–Ueda complement; consequently, only the final row of $B_n(s)$ occurs:

$$B_3(s) = \begin{pmatrix} 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}.$$

The Hirzebruch-surface block $H_n(s) \in M_4(\mathbb{Z})$ is

$$H_n(s) = \begin{pmatrix} 1 & 2 & 2s+n+4 & 2s+n+6 \\ 0 & 1 & 2s+n+2 & 2s+n+4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

With respect to $\mathcal{C}_{n,m}(s)$, the $(n+m+2) \times (n+m+2)$ matrix of the Euler form is

$$S_{n,m}(s) = \begin{pmatrix} T_n & 0 & B_n(s) \\ 0 & I_m & -\mathbf{1}_{m \times 4} \\ 0 & 0 & H_n(s) \end{pmatrix}.$$

Here $\mathbf{1}_{m \times 4}$ denotes the matrix all of whose entries are equal to 1.

Proof. On the local quotient chart $[\mathbb{A}^2/\mu_n]$, consider the standard μ_n -equivariant Koszul resolution [12, cf. Proposition 3.10]:

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}^2} \otimes \rho_{i+2} \longrightarrow (\mathcal{O}_{\mathbb{A}^2} \otimes \rho_{i+1})^{\oplus 2} \longrightarrow \mathcal{O}_{\mathbb{A}^2} \otimes \rho_i \longrightarrow \mathcal{O}_0 \otimes \rho_i \longrightarrow 0.$$

Taking equivariant Homs to $\mathcal{O}_0 \otimes \rho_j$ gives

$$\chi(\mathcal{E}_i, \mathcal{E}_j) = \delta_{ij} - 2\delta_{j,i+1} + \delta_{j,i+2},$$

which gives T_n . Although the characters are indexed cyclically, no wraparound term remains within the range $2 \leq i, j \leq n-1$; the indices in the displayed formula are therefore ordinary integers.

Let Ψ be the left adjoint of Φ . The direct specialization of the Gugiatti–Rota adjoint formula gives [9, Theorem 3.1 and Example 3.5]

$$\Psi(\mathcal{E}_i) = 0 \quad (2 \leq i \leq n-3), \quad \Psi(\mathcal{E}_{n-1}) \simeq \mathcal{O}_E(-n+1).$$

For $n \geq 4$, the remaining relevant formula is

$$\Psi(\mathcal{E}_{n-2}) \simeq \mathcal{O}_E(-n)[1].$$

When $n = 3$, the omitted source corresponds to the special representation ρ_1 .

The same divisor-duality computation as in Proposition 5.4 therefore gives, for a divisor D on $\mathbb{F}_n$,

$$\chi(\mathcal{E}_{n-2}, \Phi(\sigma^* \mathcal{O}(D))) = D \cdot C_0 + 1, \quad \chi(\mathcal{E}_{n-1}, \Phi(\sigma^* \mathcal{O}(D))) = -D \cdot C_0.$$

The first formula applies when $n \geq 4$. For the four line bundles in $\mathcal{C}_{n,m}(s)$, the intersection numbers are

$$D \cdot C_0 = 0, \quad 1, \quad s+1, \quad s+2.$$

These values give $B_n(s)$, including the stated truncation when $n = 3$. Moreover, E is disjoint from $D_1, \dots, D_m$, so

$$\chi(\mathcal{E}_i, \Phi(\mathcal{O}_{D_a}(-1))) = 0$$

for every i and a .

Because the blow-up centers are pairwise distinct, the curves $D_1, \dots, D_m$ are pairwise disjoint. Orlov’s blow-up formula [17, Theorem 4.3 and Corollary 4.4] and the divisor-duality computation in Proposition 5.4 apply to each of them and give

$$\chi(\mathcal{O}_{D_a}(-1), \mathcal{O}_{D_b}(-1)) = \delta_{ab}, \quad \chi(\mathcal{O}_{D_a}(-1), \sigma^* L) = -1, \quad \chi(\sigma^* L, \mathcal{O}_{D_a}(-1)) = 0$$

for every line bundle L on $\mathbb{F}_n$. These pairings yield I_m , $-\mathbf{1}_{m \times 4}$, and the opposite zero block.

Finally, Riemann–Roch on $\mathbb{F}_n$, using

$$K_{\mathbb{F}_n} = -2C_0 - (n+2)f$$

and

$$\chi(\mathcal{O}(A), \mathcal{O}(B)) = 1 + \frac{1}{2}(B - A) \cdot (B - A - K_{\mathbb{F}_n}),$$

gives $H_n(s)$ for the four Hille–Perling line bundles. Combining these calculations with the Ishii–Ueda ordering [12, Theorem 1.4], which supplies the remaining lower-left zero blocks, proves the formula for $S_{n,m}(s)$. $\square$

For $n = 3$ and $m = 8$, one has $T_3 = (1)$ and $B_3(s) = (0, -1, -(s+1), -(s+2))$, while $H_3(s)$ is the 4×4 block in Proposition 5.4. Together with I_8 and $-\mathbf{1}_{8 \times 4}$, the formula recovers $S_{\chi_{10}}(s)$.

In particular, Proposition 6.2 applies to the one-point Cavey–Prince cascades starting from $\mathbb{P}(1, 1, n)$ whenever the centers on the minimal resolution are pairwise distinct; see [4]. For several singularities, the initial model must already carry the prescribed basket. Cavey–Prince, for example, construct two-point models

$$X_{k_1+k_2} \subset \mathbb{P}(1, 1, k_1, k_2),$$

with singularities of types $\frac{1}{k_1}(1, 1)$ and $\frac{1}{k_2}(1, 1)$ [4, Corollary 7.1]. Extending the closed matrix formula to such families requires accounting for the separate local blocks and their interaction with the chosen blow-up configuration.

A Brauer groups and obstructions in the Corti–Heuberger cascades

We retain the notation of [Remark 4.2](#). Thus $E_1, \dots, E_k$ are the exceptional (-3) -curves, Λ is the lattice they generate, and $\theta : \text{Pic}(\tilde{X}) \rightarrow \Lambda^\vee$ is the intersection map. Since the minimal resolution is rational, that remark identifies $\text{Br}(X)$ with $\text{coker}(\theta)$. We write $e_1, \dots, e_k$ for the basis of $\Lambda^\vee$ dual to the E_i . We also retain the notation Z_k of [Section 5](#) for the chosen representative of the qG -rigid base of the k -point cascade. By [\[5, Theorem 6 and Remark 7\]](#), each Z_k is the unique isomorphism class in its base family, and [\[5, Corollary 8\]](#) expresses every member of an index-one family $X_{k,d}$ in the corresponding cascade as a sequence of blow-ups of nonsingular points from Z_k . The auxiliary families in the one- and two-point cascades are treated separately in [Appendix A.5](#). We determine the cokernel for all six cascades.

Bright’s exact sequence applies to arbitrary rational surface singularities [\[3, Proposition 1\]](#), whereas the classification carried out in [\[3, Section 3\]](#) concerns the classical singular del Pezzo case obtained by contracting (-2) -curves. In the Corti–Heuberger cascades the exceptional curves over the singular points have self-intersection -3 , so the calculations below are not contained in that classification.

A.1 Glue in the exceptional lattice

Define the saturation of Λ in $\text{Pic}(\tilde{X})$ by

$$\Lambda^{\text{sat}} = (\Lambda \otimes \mathbb{Q}) \cap \text{Pic}(\tilde{X}).$$

The quotient $\Lambda^{\text{sat}}/\Lambda$ measures the failure of the exceptional lattice to be primitive; following standard lattice terminology, we call its elements *glue classes*. The discriminant group

$$A_\Lambda := \Lambda^\vee/\Lambda$$

carries the induced discriminant bilinear form

$$b_\Lambda : A_\Lambda \times A_\Lambda \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

The standard gluing theory for integral lattices identifies integral overlattices with isotropic subgroups of the discriminant group; see [\[16, Section 1.4\]](#), using the discriminant bilinear form in the possibly odd case rather than the discriminant quadratic form of an even lattice. Hence $\Lambda^{\text{sat}}/\Lambda \subset A_\Lambda$ is isotropic for b_Λ .

Lemma A.1. *Every class in $\Lambda^{\text{sat}}/\Lambda$ can be represented in the form*

$$L = \frac{1}{3} \sum_{i=1}^k a_i E_i, \quad a_i \in \{0, \pm 1\}.$$

Such a representative satisfies the congruences

$$\sum_{i=1}^k a_i^2 \equiv 0 \pmod{3}, \quad \sum_{i=1}^k a_i \equiv 0 \pmod{3}.$$

Consequently:

- (i) *if $k = 1$ or $k = 2$, the lattice Λ is primitive;*

(ii) if $3 \leq k \leq 5$, every nonzero glue class is represented by

$$\pm \frac{E_a + E_b + E_c}{3}$$

for a triple $\{a, b, c\}$, and the glue subgroup $\Lambda^{\text{sat}}/\Lambda$ has order at most 3;

(iii) if $k = 6$, classes supported on three or on six exceptional curves are numerically possible.

Proof. Since every class in $\text{Pic}(\tilde{X})$ has integral intersection with each E_i , the saturation Λ^{sat} is contained in $\Lambda^\vee$. The Gram matrix of Λ is $-3I_k$, so $A_\Lambda \cong (\mathbb{Z}/3)^k$, which gives the stated form of L . Integrality of the intersection form on $\text{Pic}(\tilde{X})$ gives

$$L^2 = -\frac{1}{3} \sum_i a_i^2 \in \mathbb{Z},$$

and therefore $\sum_i a_i^2 \equiv 0 \pmod{3}$. Moreover, adjunction gives

$$K_{\tilde{X}} \cdot E_i = 1.$$

It follows that

$$K_{\tilde{X}} \cdot L = \frac{1}{3} \sum_i a_i \in \mathbb{Z},$$

which proves the second congruence.

The first congruence says that the number of nonzero coefficients is divisible by 3. This proves (i). If $3 \leq k \leq 5$, a nonzero class has support of size 3, and the second congruence forces its three coefficients to have the same sign. By the cited overlattice correspondence, the glue subgroup is totally isotropic for b_Λ . Two support-three classes with distinct supports have pairing

$$-\frac{\#(A \cap B)}{3} \pmod{\mathbb{Z}}.$$

For two distinct triples in a set of at most five elements, the intersection has size 1 or 2, so the two classes cannot be orthogonal. Hence all nonzero glue classes generate the same cyclic subgroup, proving (ii). For $k = 6$, supports of size 3 and 6 satisfy the first numerical condition, subject also to the stated congruence on the signs. $\square$

Thus three exceptional curves form the smallest possible support of a nonzero glue class. For $k = 1$, the primitivity conclusion recovers the computation for $\mathbb{P}(1, 1, 3)$ in [Remark 4.2](#). For $k = 2$, Λ is likewise primitive. The blow-up description used in the proof of [Corollary 2.6](#) shows that the intersection lattice $\text{Pic}(\tilde{X})$ is unimodular: this holds for $\mathbb{P}^2$ and the Hirzebruch surfaces, and a point blow-up adds an orthogonal (-1) -class. Since $\Lambda \subset \text{Pic}(\tilde{X})$ is primitive, its quotient is torsion-free. Applying $\text{Hom}(-, \mathbb{Z})$ and using the unimodularity $\text{Pic}(\tilde{X}) \cong \text{Pic}(\tilde{X})^\vee$ shows that the restriction map $\theta : \text{Pic}(\tilde{X}) \rightarrow \Lambda^\vee$ is surjective. Hence the Brauer group vanishes:

$$\text{Br}(X) = 0.$$

A.2 Behavior under cascade blow-ups

The morphisms in a Corti–Heuberger cascade are blow-ups of nonsingular points by [5, Corollary 8]. Their effect on the intersection map is elementary.

Lemma A.2. *Let $X' \rightarrow X$ be one of the blow-ups at a smooth point occurring in a Corti–Heuberger cascade. This blow-up preserves the Brauer group:*

$$\mathrm{Br}(X') \cong \mathrm{Br}(X).$$

Proof. On minimal resolutions, the induced morphism

$$\beta : \tilde{X}' \longrightarrow \tilde{X}$$

is the blow-up of a point disjoint from $E_1, \dots, E_k$. The strict transforms of these curves give natural identifications

$$\Lambda_{X'} \cong \Lambda_X, \quad \Lambda_{X'}^\vee \cong \Lambda_X^\vee.$$

If F is the new exceptional curve, the Picard group decomposes as

$$\mathrm{Pic}(\tilde{X}') = \beta^* \mathrm{Pic}(\tilde{X}) \oplus \mathbb{Z}F, \quad F \cdot E_i = 0.$$

Therefore, the intersection numbers satisfy

$$(\beta^* D + nF) \cdot E_i = D \cdot E_i$$

for every i . Under the displayed identification of the dual exceptional lattices, the image and cokernel of θ are therefore unchanged. $\square$

It is therefore enough to compute the cokernel for one convenient representative of each cascade.

A.3 The cases $k = 3, 4, 5$

The three-point base

Corti–Heuberger exhibit polygon 20 as a toric qG -degeneration in the family $X_{3,5}$; see [5, Section 7, Table 4 and Figure 11]. By the rigidity noted above, the toric special fiber is isomorphic to the chosen base $Z_3 = X_{3,5}$. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_5 \in N$ be the primitive ray generators of its fan, in counterclockwise order, with V_i the corresponding torus-invariant divisors:

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1).$$

The first three two-dimensional cones have determinant 3, while the remaining two are smooth. The primitive rays inserted to resolve the three singular cones are

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

Let E_1, E_2, E_3 be the corresponding exceptional torus-invariant divisors. The subdivision is the minimal toric resolution; see [6, Chapters 10–11] for resolutions of toric surface singularities by subdivision into primitive rays. Computing the boundary self-intersections from the fan relations as in [6, Theorem 10.4.4] gives

$$(V_1, E_1, V_2, E_2, V_3, E_3, V_4, V_5) = (0, -3, -1, -3, -1, -3, -1, 0).$$

Thus the inserted divisors are precisely the exceptional (-3) -curves. After the unimodular change of coordinates $(x, y) \mapsto (x, x + y)$, these are the rays of the polygon cited above.

The boundary intersections give

$$\theta(V_1) = e_1, \quad \theta(V_2) = e_1 + e_2, \quad \theta(V_4) = e_3.$$

Consequently, the dual basis vectors satisfy

$$e_1 = \theta(V_1), \quad e_2 = \theta(V_2) - \theta(V_1), \quad e_3 = \theta(V_4),$$

so θ is surjective and the Brauer group vanishes:

$$\mathrm{Br}(Z_3) = \mathrm{Br}(X_{3,5}) = 0.$$

Equivalently, the divisor classes

$$\mathcal{O}(-V_1), \quad \mathcal{O}(V_1 - V_2), \quad \mathcal{O}(-V_4)$$

have intersection $-\delta_{ij}$ with E_j .

For the four- and five-point calculations we use the k -gon construction. It starts with a smooth del Pezzo surface of degree k containing a cyclic configuration $C_1 + \cdots + C_k$ of (-1) -curves, blows up the vertices $C_i \cap C_{i+1}$, and contracts the strict transforms of the C_i . Corti–Heuberger show that the resulting surface has k singularities of type $\frac{1}{3}(1, 1)$, degree $k/3$, and belongs to the family $X_{k,k/3}$; see [5, Section 3.5, “ k -gons”]. Thus the models used below are respectively members of $X_{4,4/3}$ and $X_{5,5/3}$. Let F_i be the exceptional curve over the i -th vertex and E_i the strict transform of C_i , with cyclic indices. Direct intersection calculations give

$$\theta(F_i) = e_i + e_{i+1}, \quad \theta(E_i) = -3e_i.$$

The five-point base

For example, on the degree-five del Pezzo surface $\mathrm{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, with exceptional classes Q_i , the curves

$$Q_1, \quad H - Q_1 - Q_2, \quad Q_2, \quad H - Q_2 - Q_3, \quad H - Q_1 - Q_4$$

form a cyclic pentagon of (-1) -curves. The construction above therefore gives the rigid base $Z_5 = X_{5,5/3}$. The five vectors $e_i + e_{i+1}$ are the columns of

$$A_5 = I + P_5,$$

where P_5 is the cyclic permutation matrix. A direct calculation gives

$$\det(A_5) = 2.$$

Thus $\mathbb{Z}^5/A_5\mathbb{Z}^5$ has order 2. Since the image of θ also contains $3\mathbb{Z}^5$, and multiplication by 3 is invertible on this order-two quotient, we obtain

$$A_5\mathbb{Z}^5 + 3\mathbb{Z}^5 = \mathbb{Z}^5.$$

Therefore θ is surjective, and the Brauer group vanishes:

$$\mathrm{Br}(Z_5) = \mathrm{Br}(X_{5,5/3}) = 0.$$

A four-point representative

Here it is convenient to use the member $X_{4,4/3}$, obtained from the rigid base $Z_4 = X_{4,7/3}$ by one blow-up at a nonsingular point [5, Corollary 8]. By Lemma A.2, it has the same cokernel as Z_4 . The vectors $e_i + e_{i+1}$ span a rank-three subgroup, so one additional divisor class is needed. To avoid confusing the exceptional classes on a degree-four del Pezzo surface with the (-3) -curves above, write

$$S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$$

with basis $H, Q_1, \dots, Q_5$, where $H^2 = 1$, $Q_i^2 = -1$, and all other pairings between basis elements vanish. Consider

$$\begin{aligned} C_1 &= Q_1, \\ C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, \\ C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Their self-intersections are all -1 . The adjacent intersections are

$$C_1 \cdot C_2 = C_2 \cdot C_3 = C_3 \cdot C_4 = C_4 \cdot C_1 = 1,$$

and the nonadjacent intersections are

$$C_1 \cdot C_3 = 0, \quad C_2 \cdot C_4 = 2 - 1 - 1 = 0.$$

Thus their intersection graph is a 4-cycle. Moreover,

$$Q_3 \cdot C_1 = Q_3 \cdot C_2 = Q_3 \cdot C_3 = 0, \quad Q_3 \cdot C_4 = 1.$$

For general $p_1, \dots, p_5$, these intersections are transverse; in particular, Q_3 avoids the four vertices of the cycle. If π is their blow-up, the projection formula gives

$$\theta(\pi^* Q_3) = e_4.$$

It follows successively that

$$\begin{aligned} e_1 &= \theta(F_4) - e_4, \\ e_2 &= \theta(F_1) - e_1, \\ e_3 &= \theta(F_2) - e_2. \end{aligned}$$

Hence θ is surjective and the 4-gon representative $X_{4,4/3}$ has trivial Brauer group:

$$\text{Br}(X_{4,4/3}) = 0.$$

A.4 The six-point obstruction

For the rigid base $Z_6 = X_{6,2}$ of Section 5, Corti–Heuberger give the corresponding toric model in [5, Section 2.2.5]. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_6 \in N$ be the primitive ray generators of its fan, with V_i the corresponding torus-invariant divisors:

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

in counterclockwise order. Indices are cyclic modulo 6 throughout. Every adjacent cone has determinant 3, and hence defines a cyclic quotient singularity of order 3. The primitive rays inserted in the minimal toric resolution are

$$u_i = \frac{v_i + v_{i+1}}{3}, \quad i \in \mathbb{Z}/6\mathbb{Z}.$$

The inserted rays are explicitly

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Let E_i be the exceptional torus-invariant divisor corresponding to u_i . As in the three-point case, this subdivision gives the minimal toric resolution; see [6, Chapters 10–11]. The resolved boundary is the alternating cycle

$$V_1, E_1, V_2, E_2, \dots, V_6, E_6,$$

and, by [6, Theorem 10.4.4], the fan relations give

$$V_i^2 = -1, \quad E_i^2 = -3.$$

Thus the minimal resolution of each singularity consists of one (-3) -curve, so all six singularities are of type $\frac{1}{3}(1, 1)$.

We now exhibit the divisibility responsible for the obstruction. The principal-divisor formula $\operatorname{div}(\chi^m) = \sum_\rho \langle m, v_\rho \rangle D_\rho$ of [6, Theorem 4.1.3], applied to the character $m = (0, 1)$, gives

$$\begin{aligned} 0 &= E_1 + 3V_2 + 2E_2 + 3V_3 + E_3 - E_4 \\ &\quad - 3V_5 - 2E_5 - 3V_6 - E_6. \end{aligned}$$

Rearranging gives

$$E_1 - E_2 + E_3 - E_4 + E_5 - E_6 = 3L,$$

where the divisor L is

$$L = -V_2 - V_3 + V_5 + V_6 - E_2 + E_5 \in \operatorname{Pic}(\tilde{Z}_6).$$

Thus the lattice generated by $E_1, \dots, E_6$ is not primitive.

For a complete toric variety, the torus-invariant prime divisors generate $\operatorname{Cl}(\tilde{Z}_6)$, and smoothness gives $\operatorname{Cl}(\tilde{Z}_6) = \operatorname{Pic}(\tilde{Z}_6)$; see [6, Theorem 4.1.3]. Hence the toric boundary generates $\operatorname{Pic}(\tilde{Z}_6)$, and its image under θ is generated by

$$\theta(E_i) = -3e_i, \quad \theta(V_j) = e_{j-1} + e_j,$$

with cyclic indices. The matrix with columns $e_{j-1} + e_j$ has Smith normal form

$$\operatorname{diag}(1, 1, 1, 1, 1, 0).$$

Thus its image has rank five and is primitive. Each generator lies in the kernel of the primitive alternating-sum homomorphism

$$\mathbb{Z}^6 \longrightarrow \mathbb{Z}, \quad (a_1, \dots, a_6) \longmapsto a_1 - a_2 + a_3 - a_4 + a_5 - a_6.$$

Since this kernel also has rank five, the two lattices coincide. Adding the generators $-3e_i$ gives

$$\mathrm{im}(\theta) = \left\{ (a_1, \dots, a_6) \in \mathbb{Z}^6 : a_1 - a_2 + a_3 - a_4 + a_5 - a_6 \equiv 0 \pmod{3} \right\}.$$

Consequently, the cokernel and Brauer group are

$$\mathrm{coker}(\theta) \cong \mathbb{Z}/3, \quad \mathrm{Br}(Z_6) = \mathrm{Br}(X_{6,2}) \cong \mathbb{Z}/3.$$

In particular, there cannot exist divisor classes L_i satisfying

$$L_i \cdot E_j = -\delta_{ij},$$

since such classes would make θ surjective.

By [14, Corollary 4.8], a semiorthogonal decomposition of the resolution whose components are untwisted adherent to the connected components of the exceptional divisor would force $X_{6,2}$ to be torsion-free in the sense of [14, Definition 4.7], equivalently $\mathrm{Br}(X_{6,2}) = 0$. The nonzero class computed above therefore prevents such an untwisted adherent decomposition. In the expected untwisted construction, each local algebra would be $K(3,1)$. This does not rule out unrelated semiorthogonal decompositions or the twisted adherent constructions of KKS.

A.5 Proof of Theorem 1.3

The case $k = 1$ is the computation of $\mathrm{Br}(\mathbb{P}(1, 1, 3))$ in Remark 4.2, and the case $k = 2$ follows from Lemma A.1. For $k = 3$ and $k = 5$, Appendix A.3 computes directly on the rigid bases $Z_3 = X_{3,5}$ and $Z_5 = X_{5,5/3}$. For $k = 4$, it computes on the 4-gon representative $X_{4,4/3}$, which belongs to the four-point cascade and is obtained from $Z_4 = X_{4,7/3}$ by one smooth blow-up. These three intersection maps are surjective, while Appendix A.4 gives cokernel $\mathbb{Z}/3$ on the rigid base $Z_6 = X_{6,2}$. The uniqueness of the rigid bases noted at the beginning of the appendix, together with Lemma A.2 and [5, Corollary 8], propagates the computations to every member of the index-one families in the corresponding cascades.

It remains to cover the two auxiliary families. Let $B \in B_{1,16/3}$, and choose a smooth-point blow-up $X' \rightarrow B$ as in [5, Remark 9(1)]. The same remark gives an alternative sequence of four contractions of floating (-1) -curves from X' to $\mathbb{P}(1, 1, 3)$. Applying Lemma A.2 along these routes gives

$$\mathrm{Br}(B) \cong \mathrm{Br}(X') \cong \mathrm{Br}(\mathbb{P}(1, 1, 3)) = 0.$$

Similarly, let $B \in B_{2,8/3}$. By [5, Remark 9(2)], a suitable smooth-point blow-up $X' \rightarrow B$ admits an alternative contraction to $X_{2,17/3}$. Hence the Brauer groups satisfy

$$\mathrm{Br}(B) \cong \mathrm{Br}(X') \cong \mathrm{Br}(X_{2,17/3}) = 0.$$

Thus all members of all six cascades are covered, proving Theorem 1.3. $\square$

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing, organization, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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